MDS-Net: An Image-Text Enhanced Multimodal Dual-Branch Siamese Network for Remote Sensing Change Detection

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***Abstract*—Remote sensing change detection (RSCD), which aims to identify differences between bitemporal images, has made great progress through the application of deep learning methods. Convolutional neural networks and transformers are extensively employed in remote sensing image change detection, achieving promising results. However, current models predominantly focus on visual representation learning while neglecting the potential of multimodal learning methods. Consequently, this leads to issues such as inaccurate identification of nonsemantic changes, incomplete boundary extraction due to the degradation of local feature details, and the loss of small targets. Recently, a novel method for efficiently learning from natural language supervision, known as contrastive language image pretraining, has been proposed. Inspired by this work, we aim to leverage pixel-text relationships to guide the training of detection (RSCD) models. We propose a novel end-to-end multimodal dual-branch Siamese network, named MDS-Net, specifically designed for RSCD. In addition, we propose a U-Shaped vision-language guided transformer multimodal decoder to enhance the interactions between visual and textual information. To capture detailed semantic change features, we also introduce a novel difference feature enhancement. module. Extensive experiments demonstrate that our model outperforms other state-of-the-art methods, with IoU values of 85.60%, 76.81%, 89.55%, 94.51%, and 71.77% on the LEVIR-CD, LEVIR-CD+, WHUCD, CDD, and SYSU-CD datasets, respectively.**

***Index Terms*—Change detection, multimodal learning, remote sensing image, Siamese network, transformer.**

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I. Introduction

REMOTE sensing change detection (RSCD) refers to the process of identifying changes on the Earth's surface by analyzing geographically coregistered bitemporal remote sensing images. RSCD has been applied in various fields, including urban expansion investigations [1], [2], disaster assessment [3], [4], forestry and agricultural monitoring [5], [6], and other fields. In high-resolution remote sensing images, unchanged areas greatly outnumber changed areas, and lighting variations can result in class imbalance and false changes [7], [8]. This significant loss of change information negatively impacts detection accuracy. In recent years, convolutional neural networks (CNNs) and transformers have achieved significant success in various subfields of computer vision, such as image classification [9], [10], [11], semantic segmentation [12], [13], [14], and object detection [15], [16], [17]. In view of the efficiency of deep learning techniques in image feature extraction, many methods have been successfully applied to the field of RSCD.

The change detection method based on Siamese networks is the most popular baseline [18]. Most current change detection methods rely on the Siamese network structure, enabling simultaneous processing of bitemporal images. Following this structure, numerous change detection methods have emerged, achieving good results [19], [20], [21], [22], [23]. However, the backbone networks employed by these models are typically trained on single-modal large-scale datasets such as ImageNet. They rely solely on weights pretrained on these single-modal image datasets, neglecting the rich semantic information available in multimodal data, which leads to performance bottlenecks.

In contrast to conventional supervised and self-supervised pretraining methods that rely solely on images, contrastive language image pretraining (CLIP) [24] is an efficient approach for learning from natural language supervision. After pretraining, natural language is utilized to reference learned visual concepts, enabling zero-shot transfer of the model to downstream tasks. This strategy has achieved impressive results on various visual tasks with no or very limited annotations [25], [26], [27], [28]. This remarkable success inspired us to explore its ability to adapt to RSCD. However, transferring knowledge from CLIP, which primarily focuses on image-level contrastive learning, to pixel-level prediction tasks is highly challenging due to the granularity disparity between images and pixels. In addition, due to the significantly smaller data sizes of RSCD datasets compared to vision pretraining datasets, the capability of fine-tuned models is often constrained.

Compared to successful self-supervised learning methods in the natural language processing field, computer vision still relies mainly on supervised learning. This leads to limited training data for models, insufficient generalization ability, and high cost of manually labeled datasets. CLIP aims to leverage self-supervised learning methods from the natural language processing field to train more powerful multimodal models on larger datasets. This training method helps improve the zero-shot learning ability of models. Therefore, CLIP has been widely applied in various multimodal tasks, such as image classification [29], [30], text-to-image retrieval [31], [32], image generation [33], [34], and semantic segmentation [26], [35]. We explore the transfer of knowledge from large-scale vision language pretrained models to downstream RSCD tasks. However, existing dense prediction task models are constructed on the visual-language relationships embedded in the pretrained CLIP model, raising questions about their generalization capability: Are dense prediction task models solely suited for CLIP's native image encoders, such as ResNet and ViT, or can they be adapted to other pretrained image encoders?

In this article, we present a pioneering language-guided multimodal change detection framework named MDS-Net. MDS-Net transfers the knowledge from image-text contrastive learning to the change detection task by introducing a pixel-text matching mechanism. We employ the CLIP model to generate descriptive prompts derived from a taxonomy of 56 common land cover categories [36], alongside foreground and background features. By actively leveraging pretrained knowledge from both CLIP and Swin transformer, we directly train and fine-tune our model on datasets customized for RSCD. Our model further utilizes the contextual information from images to prompt the language model through a transformer module, enabling the model to better exploit pretrained knowledge by optimizing the text embeddings. In addition, we designed a difference feature enhancement (DFE) module to produce a feature map highlighting changes, encompassing rich information regarding the differences. The main contributions of this article are as follows.

1) In our knowledge base, we propose a novel end-to-end multimodal dual-branch Siamese network, termed MDS-Net, which leverages both image and text multimodal data to significantly improve the accuracy of RSCD.
2) A novel U-shaped vision language guided transformer (UVLGT) multimodal decoder is proposed to enhance the interactions between visual and textual information, while effectively capturing long-range dependencies.
3) We design a DFE module to capture robust semantic changes from bitemporal image features.
4) The proposed MDS-Net achieves state-of-the-art (SOTA) performance on the LEVIR-CD, LEVIR-CD+, CDD, WHUCD, and SYSU-CD datasets.

II. Related Work

A. Change Detection Model Based on CNNs or Transformers

The advantages of CNN-based change detection methods lie in their ability to automatically learn hierarchical features from data, adapt to various types of remote sensing images, and effectively handle complex spatial relationships. These models enhance the network's semantic representation through structural modifications, loss function optimization, and the incorporation of attention mechanisms. IFN [37] adopts a two-stream architecture that feeds into a deeply supervised difference discrimination network, leveraging attention modules to enhance boundary completeness and internal object compactness. MapsNet [38] utilizes a two-stream fully convolutional network, augmented by a multiattention module and a novel attention module, CBAM-s, along with a pixel-shuffle image fusion network for contextual information aggregation and feature fusion. CLNet [39] incorporates cross-layer blocks within a U-Net architecture to fuse multiscale features and contextual information. Despite the demonstrated ability of these methods in change detection, CNNs lack the capability to model long-range dependencies and perform global feature extraction.

CNNs excel in feature extraction and spatial modeling, while transformers enhance the capture of global context and long-range dependencies. Researchers have begun to explore the fusion of transformer architectures with Siamese networks to capitalize on their complementary strengths. BIT [40] efficiently models long-range context within bitemporal images by leveraging the transformer to relate semantic concepts in the token-based space time. ChangeFormer [41] unifies a hierarchically structured transformer encoder with a multilayer perception in a Siamese network architecture to efficiently render the multiscale long-range details required for accurate change detection. ICIF-Net [42] extracts local and global multiscale features in parallel, leveraging the potential of integrating CNN and transformer models, followed by generating a change binary graph through intrascale cross-interaction and interscale feature fusion. TransUNetCD [43] integrates transformers into an encoder-decoder architecture to model contextual information and capture rich global features. The excessive computational demands of transformers require additional storage and computational resources for model training, significantly hindering the efficiency of change detection.

These methods struggle to effectively balance accuracy and model complexity, making the generalization capability of the algorithms a challenging task. Several novel approaches have emerged to improve the models' generalization capacity. GAS-Net [44] incorporates a global attention module and a foreground awareness module to enhance contextual relationships and foreground features, respectively. CF-GCN [8] introduces a coordinate space and feature interaction graph convolutional network, leveraging the boundary perception module and the knowledge review module for boundary feature preservation and erroneous knowledge propagation mitigation. SAGNet [45] refines deep semantic features with the global semantic aggregation module, the attention fusion module, the cross-scale fusion module, and the bilateral feature fusion module for accurate change map reconstruction and noise removal. However, these methods only consider single-modal data and models, resulting in limited accuracy improvements.

B. Differential Feature Learning in RSCD

The typical approach usually combines the features of the feature map pyramids extracted at different periods through addition, concatenation, or subtraction to obtain the differential feature map [7], [8], [38], [44], [46]. However, such methods often fail to fully capture the differences in feature maps across different periods. Concatenation preserves much of the original information from bitemporal images but lacks prior knowledge of changes. In contrast, subtraction captures prior change information through feature differences but sacrifices the original information from bitemporal images. Many researchers have proposed various solutions. ADS-Net [47] incorporates an adaptive attention fusion module following the concatenation of bitemporal features and their subtraction results to enhance useful information and suppress irrelevant information. FDCNN [48] utilizes FD-Net to generate feature difference maps at different scales and depths, while FF-Net learns change detection knowledge from a small number of pixel-level training samples. CDENet [23] introduces the content difference enhancement module to process the dual features at each encoder layer, aiming to increase spatial attention on differing regions and enhance change difference features. DPCC-Net [49] extracts change features by calculating bitemporal correlation from dual perspectives, providing prior knowledge of change while preserving bitemporal features. The CDDM [7] is integrated into the STADE-CDNet framework to learn spatial and channel attention within the difference map, thus producing an enhanced difference feature map and enhancing the network's learning capability. These methods mainly concentrate on improving the representational power of feature extractors, while overlooking the learning of differential features. The spatiotemporal characteristics of bitemporal images do not interact sufficiently, leading to an incomplete extraction of change information. A single differential feature transformation fails to capture the complex change patterns. To address the above-mentioned issues, we propose a novel DFE module to enhance differences in bitemporal features. The DFE module extracts change features by learning the optimal distance metric at each scale during training. Subsequently, it combines the difference feature maps computed through subtraction to generate the final difference feature map.

C. Multimodal Representation Learning

Change detection [37], [40], object detection [50], [51], and semantic segmentation [14], [52] are fundamental tasks in computer vision, especially in fields such as remote sensing. These tasks often follow a "pretraining+fine-tuning" paradigm [53], [54], in which models are first trained on large-scale datasets to acquire general feature representations and pattern recognition abilities. Subsequently, fine-tuning is performed on smaller, task-specific datasets to tailor the model to particular applications. This approach has proven effective in improving performance, especially when labeled data are scarce. However, pretrained models derived from general datasets such as ImageNet may not entirely fulfill the demands of remote sensing tasks due to differences in data characteristics. Existing approaches may only capture image features without fully comprehending the semantic information in remote sensing images. To address these challenges, it is crucial to combine advancements in computer vision with domain-specific knowledge to develop more effective and accurate models for remote sensing image processing.

CLIP [24] represents a novel approach for learning visual representations by leveraging contrastive learning with large-scale image text pairs. This language guided pretraining yields impressive performance on visual tasks without extensive annotations. CLIP has been integrated into various dense prediction tasks, such as classification [55], [56], semantic segmentation [57], [58], and object detection [28], [30], [36], underscoring its adaptability and significant advancements across various downstream applications. However, transferring knowledge from CLIP to dense prediction task is more challenging due to the difference in granularity between image-level and pixel-level representations. The former emphasizes instance-level representations of both images and texts, while the latter relies solely on visual information at the pixel level. In addition, existing models for dense prediction tasks rely on the visual-language relationships embedded in the pretrained CLIP model, where the CLIP image encoder is pretrained using ResNet and ViT backbones. If a pretrained image encoder is used and its feature maps do not strongly correlate with the text features output by the CLIP text encoder, this raises the question of whether the text encoder can still effectively guide the backbone network in adapting to downstream tasks. Our research investigates the design of independent end-to-end multimodal encoder-decoder architectures that integrate both image and text features to enhance the accuracy of RSCD.

III. Methodology

A. Preliminaries

1) Overview of CLIP: CLIP [24] consists of two encoders: an image encoder (ResNet [59] or ViT [60]) and a text encoder (Transformer [40]), both encoders are trained from scratch on a dataset of 400 million (image, text) pairs collected from the internet to learn SOTA image representations. The alignment of image and text embeddings, generated by these independent encoders, is achieved by computing pairwise similarities between them. After pretraining, natural language is used to reference the learned visual concepts, facilitating zero-shot transfer to various downstream tasks. By utilizing the template "a photo of a [CLS]" to construct text prompts, where [CLS] is replaced with the actual class name, CLIP computes the similarity between the image and the text prompts in the embedding space. Ultimately, the class with the highest similarity score is regarded as the predicted label for the given image.

2) Overview of Swin and CSwin Transformer: To reduce the computational complexity of global self-attention, the Swin

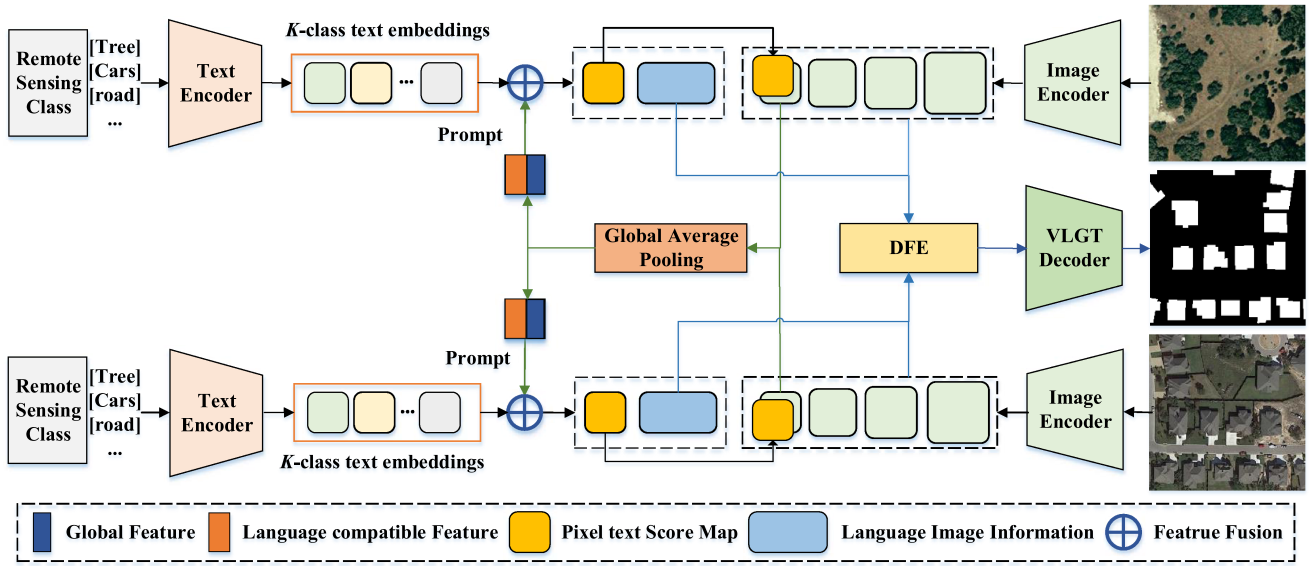

Fig. 1. Overall architecture of MDS-Net consists of several steps: First, it employs the CLIP model to generate textual representations for bitemporal images. Second, a text encoder and a transformer decoder are used to refine the text features and to identify the most relevant visual clues. Third, it calculates pixel-text score maps to guide the training of the learning process. Finally, it utilizes the DFE module to compute differential information, and subsequently employs the UVLGT multimodal decoder to enhance the interaction between visual and textual information.

transformer [11] introduces a (shifted) window-based multihead self-attention [(S)W-MSA] module that extracts long-range global features. The W-MSA and SW-MSA modules employ different window partition strategies to mitigate the effects of window locality. In addition to the W-MSA and SW-MSA modules, the Swin transformer block also includes a multilayer perceptron module, LayerNorm layers, and residual connections.

Unlike the Swin transformer, the CSWin transformer [61] introduces a cross-window self-attention mechanism by dividing input features into equal-width stripes and computing self-attention in parallel across both horizontal and vertical stripes. This parallel strategy incurs no additional computational overhead while expanding the self-attention computation area within each transformer block. To further enhance this visual transformer, CSWin also introduces locally enhanced positional encoding, which outperforms existing encoding schemes in effectively managing local positional information.

B. Overall Network Architecture

Recently, several efforts have been made to incorporate the prompts engineering techniques from the NLP community [62] to improve the adaptation of CLIP models to downstream visual tasks. We propose MDS-Net, a language-guided multimodal change detection framework that effectively leverages language priors from CLIP pretrained models. As illustrated in Fig. 1, the input of MDS-Net comprises two sets of parallel image encoders and text encoders. The feature maps generated by the new backbone do not exhibit a strong correlation with the text features produced by the CLIP text encoder, in contrast to the original image encoders (ResNet and ViT). We leverage the unsupervised prediction capability of CLIP to formulate original text prompts for bitemporal images while employing a hierarchical Swin transformer to produce multiscale features. The architecture consists of three main modules: a multimodal encoder, a DFE module, and a UVLGT decoder.

C. Multimodal Encoder

1) Textual Prior Knowledge: Due to the lack of pixel-level change detection datasets that provide text prompt labels, we generate text prompts for the bitemporal images based on the 56 common land cover categories [36], leveraging CLIP's unsupervised prediction capability. The categories predicted by CLIP for each image exhibit varying levels of confidence; thus, we utilize textual information that corresponds to these different confidence levels. Higher confidence indicates a higher probability that the image contains the corresponding category. The changing objects can be directly identified through the bitemporal text prompts, which also provide essential prior knowledge for remote sensing. This textual prior knowledge, combined with the pretrained image features, is fed into our model for training and fine-tuning.

2) Image Semantic Feature: As shown in Fig. 1, given an input bitemporal image with a resolution of $H{\times}W{\times}3$, the transformer encoder outputs feature maps X_i with resolutions of $\frac{H}{2^{i+1}} \times \frac{W}{2^{i+1}} \times C_i$, where $i = \{12,34\}$ and $C_i +1 > C_i$. Subsequently, we integrate the high-level semantic feature maps with text sequence features. A global average pooling module is employed at the end of the Swin transformer encoder to conduct visual embedding on the high-level feature maps, thereby constructing the image sequence features. The detailed calculation is given as

$$X' = \mathrm{Concat}(\mathrm{Mean}(\mathrm{Reshape}(X_4)), X_4) \quad (1)$$

$$X'' = \mathrm{Reshape}(X'W) \quad (2)$$

$$\bar{Z} = X''[:,:,0] \quad (3)$$

$$Z = \mathrm{Reshape}(X''[:,:,1:]) \quad (4)$$

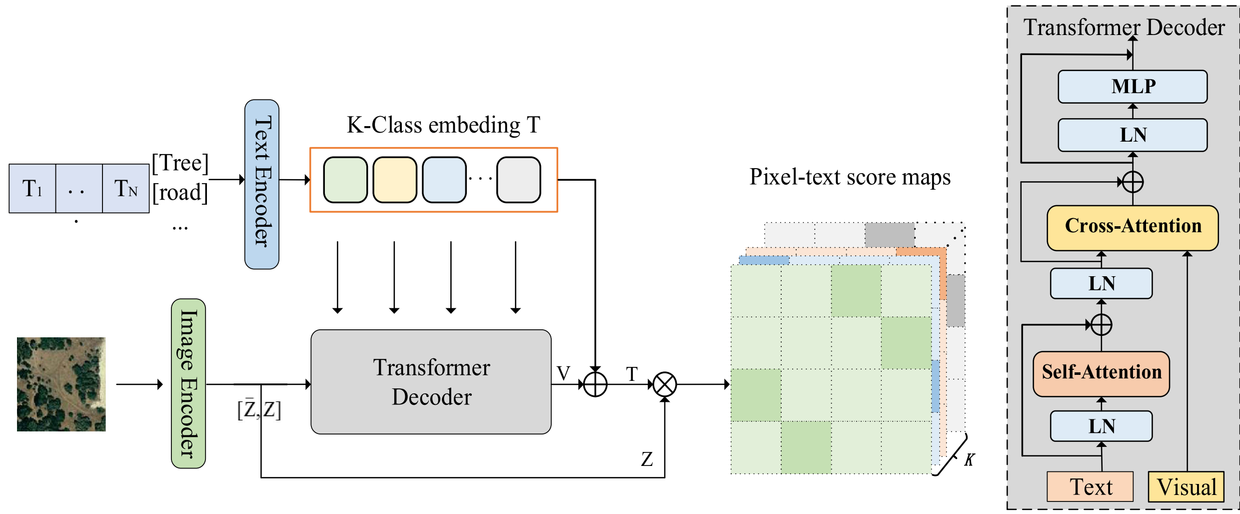

Fig. 2. Multimodal fusion transformer decoder for image and text integration.

where $X_4 \in R^{N\times C\times H_4\times W_4}$ represents the high-level feature maps extracted by the Swin transformer during the encoding stage, which contain high-level semantic information. N, C, H_4, and W_4 represent the four dimensions of X_4, namely batch size, channels, height, and width, respectively. $W \in R^{C\times C'}$ is a learnable parameter matrix, and Mean refers to global average pooling. $\bar{Z} \in \mathrm{R}^{N\times C'}$ represents the global representation, which contains rich semantic category information of the original input image. $Z \in \mathrm{R}^{N\times C'\times H_4\times W_4}$ is a language-compatible feature map that can be integrated with subsequent text sequence features.

3) Multimodal Fusion: To obtain text features, we construct text prompts using the template "a photo of a [CLS]" with K class names and utilize CLIP to extract text sequence features as $T \in R^{N\times 2\times C'}$. As shown in Fig. 2, we employ a transformer-based text encoder to refine the text feature T, which is then directly utilized as the queries for the Transformer decoder:

$$V = \mathrm{TransDecoder}(T, [\bar{\mathrm{Z}}, \mathrm{Z}]). \tag{5}$$

Here, we compute attention [63] using the following formula:

$$\mathrm{Attention}(Q, K, V) = \mathrm{Softmax}(QK^T/\sqrt{d_k})V. \tag{6}$$

The key distinction is that, for self-attention, the input consists solely of textual features, while for cross-attention, the textual features act as the query, and the visual features serve as both the key and value.

This implementation enables text features to identify the most relevant visual cues. We then update the text features using a residual connection

$$T = T + \gamma V \tag{7}$$

where γ is a learnable parameter that regulates the scaling of the residual.

T is then combined with the language-compatible feature map Z to compute the pixel-text score maps, which are employed to guide the learning process of change detection models. The detailed calculation is given as

$$S = \tilde{Z}\tilde{T}, S \in R^{N\times 2\times H_4\times W_4} \tag{8}$$

where S represents the score maps. $\tilde{Z}$ and $\tilde{T}$ are the L_2 normalized versions of Z and T along the channel dimension. $\tilde{T}$ will be integrated into the decoder to complement the semantic features during the decoding phase, enhancing the overall feature representation. The score maps S depict the outcomes of pixel-text matching, constituting one of the most critical components in our multimodal encoder. We can concatenate the score maps with the last feature map to explicitly incorporate language priors as follows:

$$X_4 = [X_4, \mathrm{S}] \in \mathrm{R}^{N\times(C'+2)\times H_4\times W_4}. \tag{9}$$

This supplements the text-related semantic information in the high-level semantic features, making the high-level semantic feature maps more discriminative.

D. Difference Feature Enhancement Module

We utilize four difference modules to compute the differences among multilevel features extracted from paired images by the hierarchical transformer encoder, aiming to learn the optimal distance (Optdist) metric at each scale. This process is illustrated by the following equation:

$$D_{\mathrm{i}}^1 = \mathrm{BN}\left(\mathrm{ReLU}\left(\mathrm{Conv2D}\left([X_{i1}, X_{i2}]\right)\right)\right) \tag{10}$$

$$D_5^1 = \mathrm{Linear}(\mathrm{Cat}(D_1^1, D_2^1, D_3^1, D_4^1)) \tag{11}$$

where X_{i1} and X_{i2} represent the features of the prechange and postchange images from the ith layer in the encoding stage, $i \in \{1, 2, 3, 4\}$. BN denotes BatchNorm, Conv2D refers to a 3×3 depth-wise convolution, and Cat denotes concatenation. As shown in Fig. 3, the detailed structure of the U-shaped visual-language guided transformer multimodal decoder is presented. We obtained $D_1^1, D_2^1, D_3^1, D_4^1$, and D_5^1, with spatial resolutions of 8×8, 16×16, 32×32, 64×64, and 128×128, respectively.

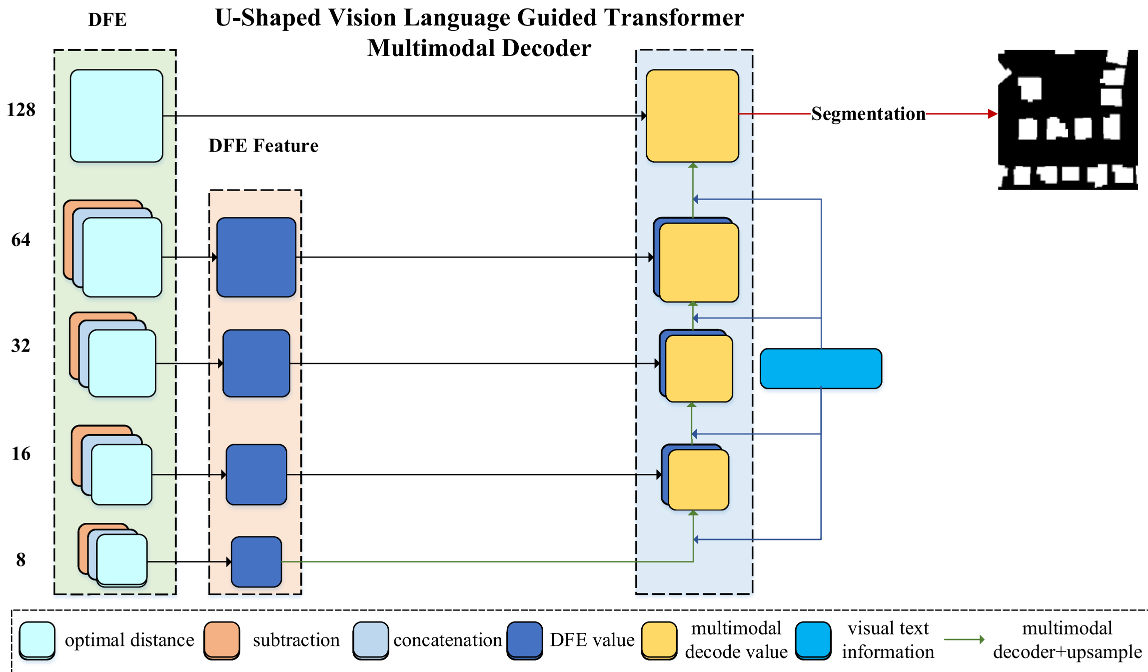

Fig. 3. Detailed structure of the U-shaped visual-language guided transformer multimodal decoder.

Second, we subtract the bitemporal features to obtain the initial difference, which is defined as follows:

$$D_i^2 = \text{Conv}\left(|X_{i1} - X_{i2}|\right). \tag{12}$$

Third, we concatenate the feature maps to preserve the complete bitemporal feature information, which is defined as follows:

$$D_i^3 = [X_{i1}, X_{i2}]. \tag{13}$$

Finally, the calculation of DFE can be described as follows:

$$D = \left[\text{CA}\left(\left[D_i^1, D_i^2, D_i^3\right]\right), D_5^1\right] \tag{14}$$

where $|\cdot|$ represents the absolute value operation, Conv is the convolution block, $[\cdot]$ denotes the tensor concatenation, and CA is the channel attention module. Our DFE module optimizes the difference feature learning strategies, significantly enhancing accuracy and robustness in the RSCD task.

E. U-Shaped Vision Language Guided Transformer Multimodal Decoder

We develop a novel decoder called the UVLGT multimodal decoder, which enhances the interaction between visual and textual information and establishes global attention relationships. The vision-language features obtained from the encoding stage encapsulate rich semantic information derived from both images and text. By integrating this semantic information into the decoding stage, our model can more effectively identify changed areas, thereby enhancing the model's feature representation capability. The calculation steps are as follows:

$$T' = \text{Concat}(\widetilde{T^1} \cdot W', \widetilde{T^2} \cdot W') \tag{15}$$

where $\widetilde{T^1}, \widetilde{T^2} \in \mathrm{R}^{\mathrm{n} \times 2 \times C'}$ represent the text sequence features extracted from the bitemporal image during the encoding stage, $W' \in R^{C' \times d}$ is a linear transformation matrix. The cross-attention mechanism [63], [61] can be employed to capture the interactions between vision and language as follows:

$$Q = W_D \cdot D', K = W_T \cdot T', V = W_V \cdot T' \tag{16}$$

$$X = \text{Softmax}(QK^T / \sqrt{d_k})V \tag{17}$$

where $D' \in R^{\mathrm{n} \times l_1 \times d}$ represents the DFE feature maps, $T' \in R^{\mathrm{n} \times l_2 \times d}$ represents the vision language feature sequence, $W_D, W_T \in R^{\mathrm{d} \times d_K}$, and $W_V \in R^{\mathrm{d} \times d_V}$ are the three linear transformation matrices. Then, X is evenly divided into nonoverlapping horizontal and vertical stripes, each with a width of *sw*. The K heads are equally divided into two parallel groups. One group applies horizontal strip self-attention, while the other group applies vertical strip self-attention. The outputs of these two groups are then concatenated. The detailed calculation is given in the following equations:

$$\text{head}_k = \begin{cases} \text{H}-\text{Attention}_k(X) k=1,\ldots,K/2 \\ \text{V}-\text{Attention}_k(X) k=K/2+1,\ldots,K \end{cases} \tag{18}$$

$$\text{CSWAtt}(X) = \text{Concat}(\text{head}_1, \ldots \text{head}_k) W^O \tag{19}$$

where H-Attention is the horizontal strip self-attention, V-Attention is the vertical strip self-attention, and $W^O \in R^{d \times d}$ is the projection matrix that projects the self-attention results into the target output dimension. The stripe width *sw* is set to 1, 4, 4, and 4 for the four stages by default. Equipped with the above self-attention mechanism, the multimodal transformer decoder is formally defined as

$$X^l = \text{MLP}(\text{LN}(\text{CSWAtt}(\text{LN}(X^{l-1})) + X^{l-1})) \tag{20}$$

where X^l denotes the output of lth transformer decoder.

In computer vision, it is widely accepted that high-level image features effectively capture the semantic information of images, while low-level image features excel at capturing the finer details. Inspired by the U-Net [13] series, we integrate multiple skip connections between the DFE and transformer decoder pathways to preserve the transmission of local details and enhance multiscale representations. Moreover, we perform multimodal decoding on different features of bitemporal remote sensing images utilizing multilevel feature fusion modules for processing, as shown in Fig. 3. The calculation steps are as follows:

$$F_4 = \mathrm{Conv}(\mathrm{Concat}(\mathrm{TransDecoder}(D_2, \mathrm{text}), D_1)) \quad (21)$$

$$F_3 = \mathrm{Conv}(\mathrm{Concat}(\mathrm{TransDecoder}(F_4, \mathrm{text}), D_3)) \quad (22)$$

$$F_2 = \mathrm{Conv}(\mathrm{Concat}(\mathrm{TransDecoder}(F_3, \mathrm{text}), D_4)) \quad (23)$$

$$F_1 = \mathrm{Conv}(\mathrm{Concat}(\mathrm{TransDecoder}(F_2, \mathrm{text}), D_5)) \quad (24)$$

where D_1, D_2, D_3, D_4, and D_5 are the different feature maps achieved by the DFE module, and text is the vision-language feature sequence.

IV. Experiments

In this section, we evaluate the performance of our proposed model by comparing it with other approaches on five datasets (LEVIR-CD, LEVIR-CD+, WHUCD, CDD, and SYSU-CD). In addition, we conduct a series of ablation experiments under different settings and frameworks.

A. Dataset Description

1) LEVIR-CD: The LEVIR-CD dataset [64] is a large-scale RSCD dataset derived from extremely high-resolution Google Earth image pairs. It comprises 31 333 fully annotated individual instances of change in buildings. The annotations in the LEVIR-CD dataset identify changes from grass, soil, ground, or areas under construction to buildings. To facilitate model training and testing, we divided the images into 10 192 nonoverlapping patches of size 256 × 256. This partitioning results in 7120, 1024, and 2048 images in the training, validation, and testing sets, respectively.

2) LEVIR-CD+: The LEVIR-CD+ dataset is an extension of the LEVIR-CD [64] dataset, encompassing 637 and 985 pairs of remote sensing images and approximately 80 000 building instances. We adopted the same cropping approach used for the LEVIR-CD dataset when processing the LEVIR-CD+ dataset. Specifically, the numbers of slices in the training, validation, and test sets are 10 192, 3172, and 1856, respectively.

3) WHUCD: The WHUCD [65] dataset is a publicly available dataset intended for building change detection tasks. It consists of a pair of aerial images, each with a resolution of 0.2 m/pixel and dimensions of 32 507 × 15 354 pixels. This dataset records changes in buildings located near Christchurch, New Zealand. We preprocess the large-scale image pairs by cropping them into nonoverlapping patch pairs of size 256 × 256 pixels. Subsequently, we partition these patch pairs into training, validation, and testing sets, comprising 5204, 743, and 1487 samples, respectively.

4) CDD: The CDD [66] dataset contains 11 pairs of season-varying Google Earth images, including seven pairs with dimensions of 4725 × 2700 pixels and four pairs with dimensions of 1900 × 1000 pixels. The spatial resolution of the dataset ranges from 0.03 to 1 m/pixel. This dataset records changes in various objects, such as buildings, roads, and vehicles, while ignoring changes caused by seasonal differences and brightness variations. The dataset is partitioned into 256 × 256 patches, comprising 10 000 training patches, 3000 validation patches, and 3000 test patches.

5) SYSU-CD: The SYSU-CD [67] dataset contains 20 000 a aerial image pairs with a spatial size of 256 × 256 pixels and a resolution of 0.5 meters that were captured in Hong Kong. The dataset includes various change types, such as vessels, roads, and vegetation. The data are divided into training, validation, and test sets at a ratio of 6:2:2.

B. Experimental Details

All experiments were conducted on an Ubuntu system equipped with an NVIDIA Tesla A40 GPU with 48 GB of memory. The implementation of all the programs are based on the mmsegmentation [68] codebases. We employ the AdamW optimizer with an initial learning rate of 6*e*-5, a weight decay of 0.01, and a total of 40*k* iterations. Learning rate warmup is conducted for the first 1500 iterations, followed by linear decay. The batch size in the referenced studies is generally set to 16. Thus, to ensure a fair comparison, we also set our batch size to 16. We adopted the default augmentation settings provided by mmsegmentation, comprising random flipping, random rotation, random cropping, and random photometric distortion. The parameters of all the models are initialized with officially provided pretrained weights. All models are trained with an input size of 256 × 256, utilizing the soft cross-entropy loss function.

To generate text prompts corresponding to remote sensing images, initial inference is conducted using the CLIP model on the original image slices extracted from the dataset. The inference results for the 56 predefined categories are stored in a JSON file. For language domain prompting, a context length of 64 is employed. During the training phase, if the encoding length surpasses the context length, the text is truncated to 64 characters. Then, we actively leverage the language priors from CLIP pretrained models and the image priors from the Swin transformer pretrained on ImageNet. By leveraging the knowledge acquired from large-scale pretraining, we train and fine-tune our model on datasets specifically customized for RSCD.

C. Evaluation Metrics

Following the widely used evaluation protocol in change detection studies, our experiments employ five typical indicators: overall accuracy (OA), intersection over union (IoU), precision (Prec), recall (Rec), and F1 score (F1). Formally, five metrics are defined as follows:

$$\mathrm{OA} = \mathrm{TP} + \mathrm{TN} / (\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}) \quad (25)$$

TABLE I
COMPARISONS OF DETECTION PERFORMANCE ON LEVIR-CD DATASET

Model	Backbone	OA	IoU	F1	Prec	Rec
BIT[40]	Resnet18	98.92	80.66	89.29	90.56	88.06
ChangeFormer[41]	Transformer	99.04	82.42	90.36	92.13	88.66
ICIF-NET[42]	Resnet18	99.02	82.21	90.24	91.34	89.16
DMINet[69]	Resnet18	99.05	82.60	90.47	92.07	88.93
GAS-Net[44]	Resnet18	99.08	83.47	90.45	91.55	**90.99**
CF-GCN[8]	Resnet50	99.06	82.83	90.61	91.76	89.48
STADE-CDNet[7]	Resnet18	98.92	80.88	89.43	88.75	90.12
ChangeCLIP(RN50)[36]	Resnet50	<u>99.18</u>	<u>84.90</u>	<u>91.83</u>	<u>93.24</u>	90.47
TransUNetCD*[43]	Resnet50	-	83.67	91.11	92.43	89.82
SaDL_CD*[20]	Resnet18	-	79.75	88.74	90.91	86.66
CDENet*[23]	U-Net	98.84	79.27	88.44	89.75	87.16
DPCC-Net*[49]	Resnet50	-	81.85	90.02	90.63	89.41
MDS-Net(ViT-B/16)	**Swin-T**	**99.22**	**85.60**	**92.24**	**93.60**	<u>90.92</u>

$$\mathrm{IoU} = \mathrm{TP}/\left(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}\right) \tag{26}$$

$$\mathrm{Prec} = \mathrm{TP}/\left(\mathrm{TP}+\mathrm{FP}\right) \tag{27}$$

$$\mathrm{Rec} = \mathrm{TP}/\left(\mathrm{TP}+\mathrm{FN}\right) \tag{28}$$

$$\mathrm{F1} = 2/\left(\mathrm{Prec}^{-1}+\mathrm{Rec}^{-1}\right) \tag{29}$$

where TP, FP, TN, and FN represent the numbers of true positives, false positives, true negatives, and false negatives, respectively.

D. Benchmark Methods

To validate the effectiveness of our proposed MDS-Net, we compared it with several representative single-modal change detection methods, including STADE-CDNet [7], CF-GCN [8], ICIF-NET [42], DMINet [69], GAS-Net [44], ChangeFormer [41], BIT-CD [40], TransUNetCD [43], SaDL [20], DPCC-Net [49], and ChangeCLIP [36]. These methods, proposed in recent years, represent the SOTA in change detection and encompass a variety of architectures and techniques. Our MDS-Net pioneers a novel multimodal vision language framework specifically tailored for RSCD, ingeniously integrating visual and textual data from the encoder to the decoder. By further integrating the DFE module, our model enhances the learning and semantic understanding of change-related features, consequently achieving SOTA performance. Our comparative analysis confirms that MDS-Net surpasses existing models in the task of RSCD.

E. Quantification and Visualization of the Results

To evaluate the effectiveness of the proposed MDS-Net model, we conducted extensive experiments on five bitemporal RSCD datasets: LEVIR-CD, LEVIR-CD+, WHUCD, CDD, and SYSU-CD. According to the source code provided in the papers of the compared methods, we retrained some models on these five datasets for comparison. A high precision indicates a low commission error, while a high recall indicates a low omission error. The F1, IoU, and OA metrics on the foreground change class, where high values correspond to high detection accuracy. The data marked with "$*$" are taken directly from the original paper, while the symbol "-" indicates data not available in the original paper. As shown in Tables I–V, the proposed MDS-Net achieves significant improvements in all accuracy metrics on the five RSCD datasets. The values in bold and underlined indicate the best and second-best performance in each column, respectively.

1) Building Change Detection: This section presents the results of MDS-Net on detecting building changes on the LEVIR-CD, LEVIR-CD+, and WHUCD datasets. The LEVIR-CD dataset is used to verify the performance of detecting small and dense building changes, while the WHUCD dataset is used to detect large and sparse building changes. As shown in Table I, the proposed MDS-Net achieves the best performance on the LEVIR-CD dataset with the highest OA of 99.22%, IoU of 85.60%, F1 score of 92.24%, and precision of 93.60%. Its recall is 90.92%, which is slightly lower than that of GAS-Net (90.99%). Table II presents the accuracy comparison results on the LEVIR-CD+ dataset, demonstrating that MDS-Net attains the highest OA (99.87%), IoU (76.81%), F1 score (86.89%), precision (88.51%), and recall (85.32%) on this dataset, surpassing other models by a large margin. Table III lists the change detection results for the WHUCD dataset, showing that the proposed MDS-Net achieves the best performance with the highest OA (99.54%), IoU (89.55%), F1 score (94.49%), precision (96.38%), and recall (92.66%). In addition, on these three datasets, the recall of MDS-Net is much greater than that of the other methods, indicating a high rate of true positive (TP) detections and a low incidence of false negatives. These results indicate its superior ability to detect small changes and further reflect its strong generalizability and robustness.

In Figs. 4–6, qualitative analyses further illustrate the change maps of the test sets from the LEVIR-CD, LEVIR-CD+, and WHUCD datasets, with distinct colors indicating detection accuracy: TPs marked in white, true negatives (TN) in black,

TABLE II
COMPARISONS OF DETECTION PERFORMANCE ON LEVIR-CD+ DATASET

Model	Backbone	OA	IoU	F1	Prec	Rec
BIT[40]	Resnet18	98.42	68.86	81.56	83.21	79.97
ChangeFormer[41]	Transformer	98.50	70.29	82.56	99.11	80.62
ICIF-NET[42]	Resnet18	98.61	72.37	83.97	84.62	83.34
DMINet[69]	Resnet18	98.69	73.44	84.69	<u>86.69</u>	82.78
GAS-Net[44]	Resnet18	98.20	65.98	79.18	79.84	79.51
CF-GCN[8]	Resnet50	98.60	70.95	83.01	88.34	78.28
STADE-CDNet[7]	Resnet18	98.46	69.81	82.22	83.24	81.24
ChangeCLIP(RN50)[36]	Resnet50	<u>98.75</u>	<u>74.81</u>	<u>85.59</u>	86.65	<u>84.55</u>
MDS-Net(RN50)	**Swin-T**	**98.87**	**76.81**	**86.89**	**88.51**	**85.32**

TABLE III
COMPARISONS OF DETECTION PERFORMANCE ON WHUCD DATASET

Model	Backbone	OA	IoU	F1	Prec	Rec
BIT[40]	Resnet18	98.28	68.11	81.03	76.55	86.06
ChangeFormer[41]	Transformer	98.97	77.51	87.33	91.87	83.22
ICIF-NET[42]	Resnet18	98.99	78.22	87.78	90.57	85.16
DMINet[69]	Resnet18	98.44	70.47	82.68	78.62	87.18
GAS-Net[44]	Resnet18	98.93	78.37	85.15	90.78	87.87
CF-GCN[8]	Resnet50	99.10	80.22	89.03	92.83	85.52
STADE-CDNet[7]	Resnet18	98.33	69.53	82.03	75.91	89.23
ChangeCLIP(ViT-B/16)[36]	ViT-B/16	<u>99.42</u>	<u>86.99</u>	93.04	<u>95.86</u>	<u>90.38</u>
TransUNetCD*[43]	Resnet50	-	84.42	<u>93.59</u>	93.59	89.60
SaDL_CD*[20]	Resnet18	-	81.78	89.98	93.10	87.05
MDS-Net(ViT-B/16)	**Swin-T**	**99.54**	**89.55**	**94.49**	**96.38**	**92.66**

TABLE IV
COMPARISONS OF DETECTION PERFORMANCE ON CDD DATASET

Model	Backbone	OA	IoU	F1	Prec	Rec
BIT[40]	Resnet18	97.05	76.66	86.79	92.07	82.09
ChangeFormer[41]	Transformer	98.71	89.57	94.60	94.77	94.24
ICIF-NET[42]	Resnet18	98.96	91.53	95.58	95.82	95.34
DMINet[69]	Resnet18	99.21	93.54	96.66	<u>96.87</u>	96.46
GAS-Net[44]	Resnet18	98.94	91.71	96.21	95.16	95.68
CF-GCN[8]	Resnet50	99.08	92.49	96.10	96.56	95.64
STADE-CDNet[7]	Resnet18	96.90	74.02	85.07	96.76	74.99
ChangeCLIP(RN50)[36]	Resnet50	<u>99.23</u>	<u>93.72</u>	<u>96.76</u>	96.72	<u>96.79</u>
MDS-Net(RN50)	**Swin-T**	**99.33**	**94.51**	**97.18**	**97.03**	**97.32**

false-positives (FP) in red, and false-negatives (FN) in green. MDS-Net accurately detects nearly all the changed buildings across the three datasets, effectively identifying the regions and edges of these buildings even under the challenges posed by building shadows and dense distributions, while the other methods tend to misclassify more unchanged pixels as changed pixels, leading to inaccuracies in the delineation of building boundaries. In conclusion, our MDS-Net network outperforms all competing methods, demonstrating the effectiveness of our proposed approach.

2) Intraclass Change Detection: This section presents the comparative results of MDS-Net and other change detection models on the intraclass change detection task using the CDD and SYSU-CD datasets. The intraclass change detection datasets cover various categories of changing objects. Notably, the CDD dataset excludes changes resulting from seasonal differences and

TABLE V
COMPARISONS OF DETECTION PERFORMANCE ON SYSU-CD DATASET

Model	Backbone	OA	IoU	F1	Prec	Rec
BIT[40]	Resnet18	90.11	63.09	77.37	84.03	71.69
ChangeFormer[41]	Transformer	90.34	63.40	77.60	85.64	70.94
ICIF-NET[42]	Resnet18	89.44	62.27	76.75	79.81	73.92
DMINet[69]	Resnet18	91.05	66.21	79.67	85.82	74.35
GAS-Net[44]	Resnet18	89.24	62.86	77.18	77.21	77.20
CF-GCN[8]	Resnet50	91.24	66.17	79.64	**88.13**	72.64
STADE-CDNet[7]	Resnet18	86.68	57.11	72.70	70.85	74.65
ChangeCLIP(ViT-B/16)[36]	ViT-B/16	92.39	71.10	83.11	86.77	**79.74**
CDENet*[23]	U-Net	89.66	61.19	75.93	84.22	69.12
DPCC-Net*[49]	Resnet50	-	66.63	79.97	81.05	78.93
MDS-Net(ViT-B/16)	**Swin-T**	**92.62**	**71.77**	**83.57**	88.02	79.54

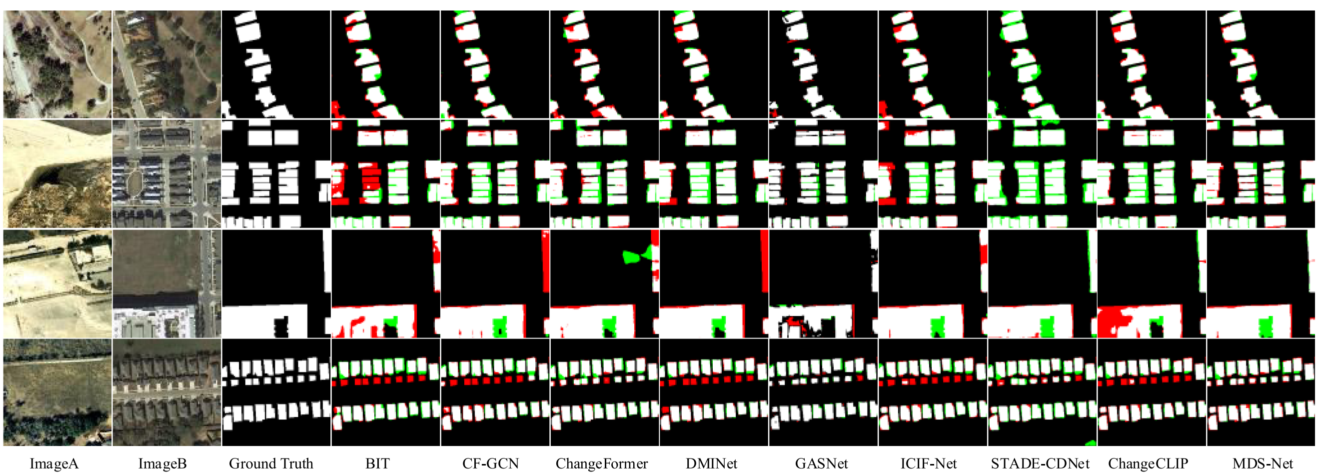

Fig. 4. Visualization results of different models on the LEVIR-CD. White, black, green, and red represent TP, TN, FP, and FN, respectively.

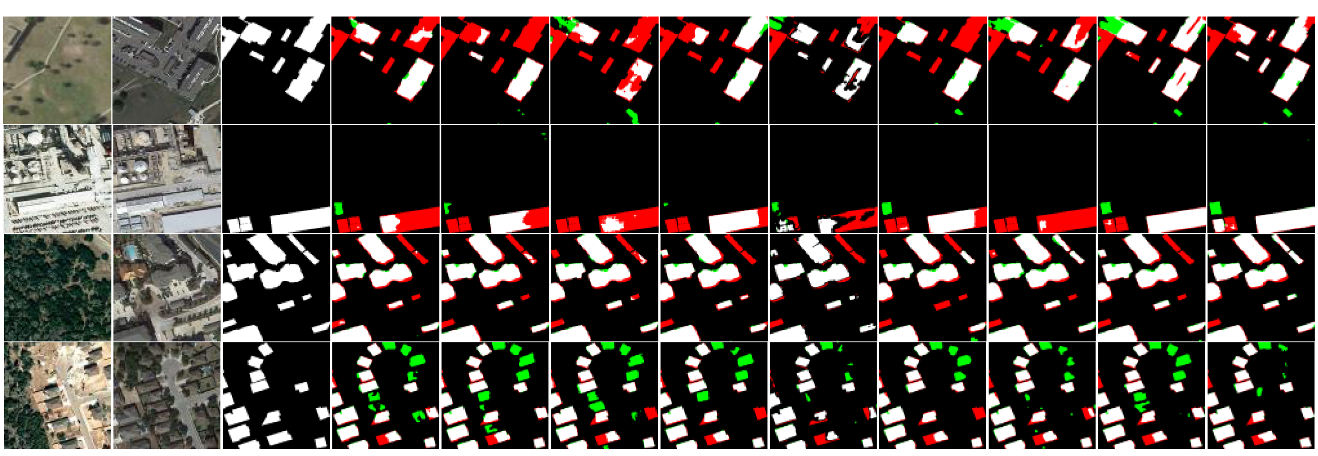

Fig. 5. Visualization results of different models on the LEVIR-CD+. White, black, green, and red represent TP, TN, FP, and FN, respectively.

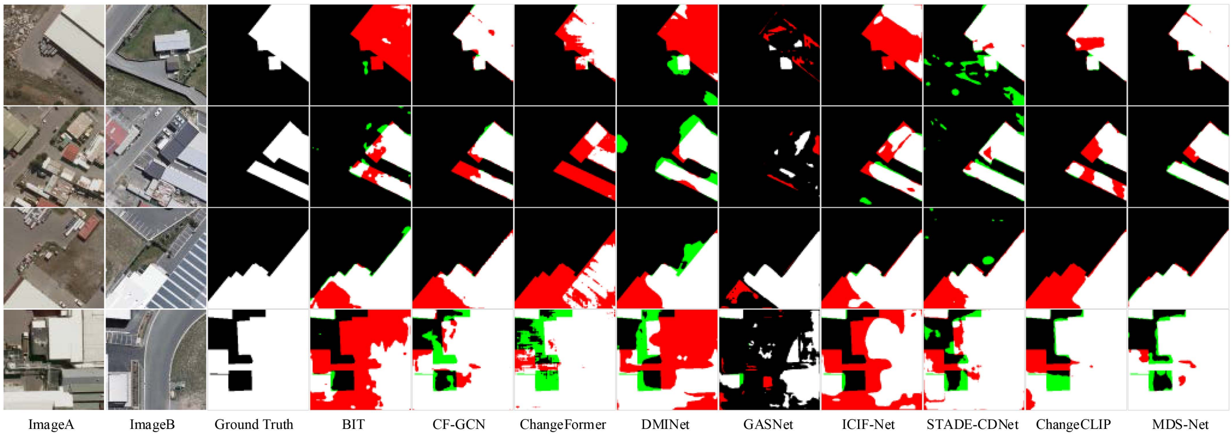

Fig. 6. Visualization results of different models on the WHUCD. White, black, green, and red represent TP, TN, FP, and FN, respectively.

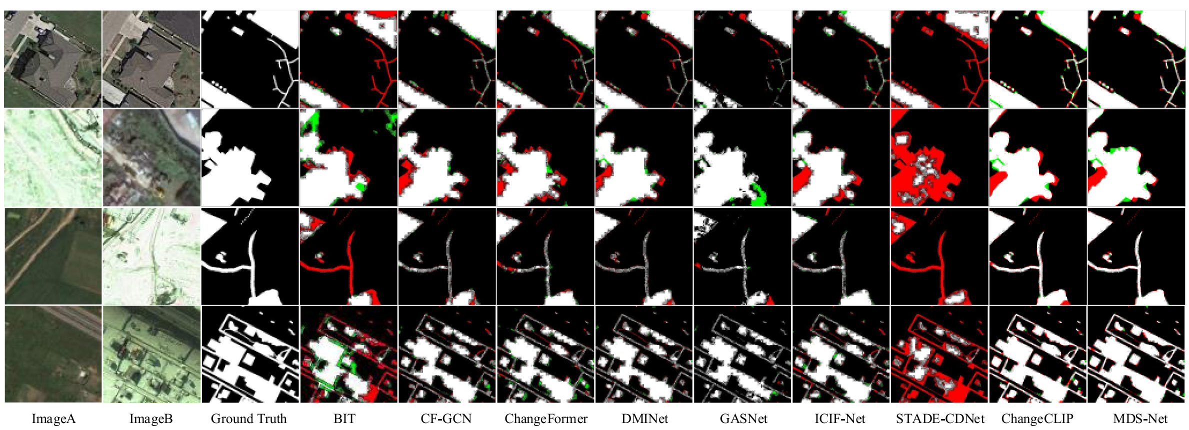

Fig. 7. Visualization results of different models on the CDD. White, black, green, and red represent TP, TN, FP, and FN, respectively.

brightness, presenting challenges for change detection. Table IV presents the experimental results on the test set of the CDD dataset. It is evident that our model achieves significantly superior performance across all five metrics compared to the other methods, with values of 99.33%, 94.51%, 97.18%, 97.03%, and 97.32%. Table V presents the change detection results for the SYSU-CD dataset, where MDS-Net outperforms almost all the compared methods across all the indicators. Due to the vague semantic information of changed objects in the SYSU-CD dataset, which encompasses objects of multiple categories, other change detection methods struggle to accurately identify these objects. In contrast, MDS-Net employs a multimodal vision language approach to accurately detect changed objects of all categories, enabling it to identify the changed areas of all objects effectively. The accuracy comparison results indicate that MDS-Net achieves the highest OA (92.62%), IoU (71.71%), and F1 (83.57%) on this dataset. The precision (88.02%) and recall (79.54%) are slightly lower, yet the overall detection performance surpasses that of all other models.

The inference results of the test sets from the CDD and SYSU-CD datasets are depicted in Figs. 7 and 8. These results demonstrate the exceptional performance of MDS-Net on both datasets. Despite the significant interference from illumination and seasonal changes in CDD, as well as the ambiguous semantic information of changed objects in SYSU-CD, MDS-Net effectively identifies various types of changed objects with binary labels.

F. Ablation Study

To validate the effectiveness and superiority of MDS-Net, we performed extensive ablation studies involving a multimodal encoder, a DFE module, and a UVLGT multimodal decoder across five change detection datasets with different scenarios. In addition, we utilized CLIP's ViT-B/16 model to infer text prompts corresponding to remote sensing images by processing sliced original images extracted from the dataset. The baseline model employs only the Swin transformer backbone as the image encoder, obtains differential features via conventional subtraction, and utilizes the cross-shaped window transformer (CSWT) [70] in the decoder, providing a performance benchmark for single-modal models.

As shown in Tables VI–X, the proposed multimodal encoder, DFE module, and UVLGT multimodal decoder further optimize the performance of the baseline network, demonstrating the value of each innovation. The DFE module outperforms conventional subtraction in extracting differential features, exhibiting superior representational power for these features. Furthermore, based on MDS-Net+DFE, the UVLGT integrates textual and

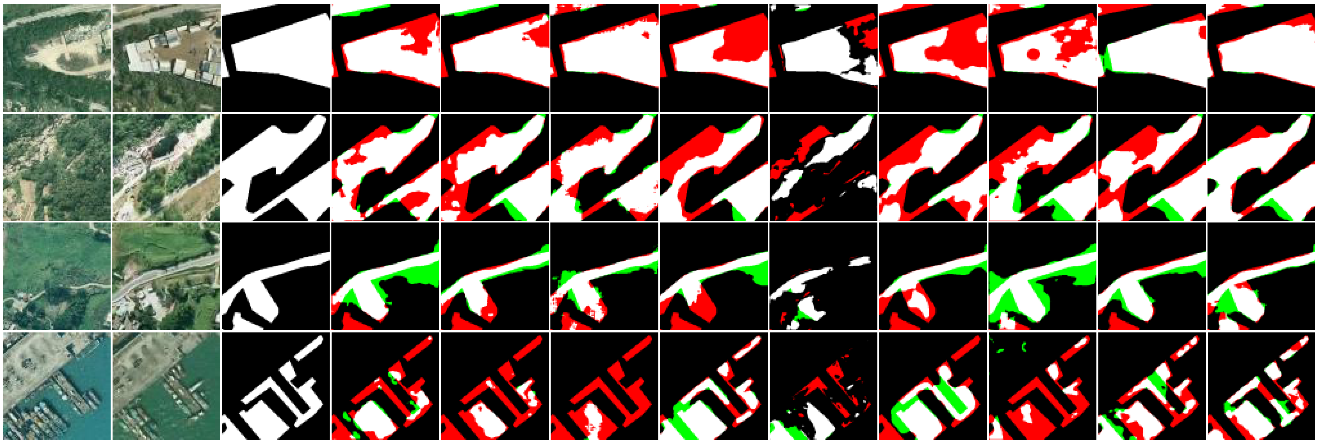

Fig. 8. Visualization results of different models on the SYSU-CD. White, black, green, and red represent TP, TN, FP, and FN, respectively.

TABLE VI
ABLATION EXPERIMENTS OF EACH COMPONENT OF THE MDS-NET ON THE LEVIR-CD DATASET

Method	Decoder	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
Baseline + Sub	CSWT	99.16	84.49	91.59	93.25	90.00
Baseline + DFE	CSWT	99.20	85.19	92.00	93.25	90.79
MDS-Net + Sub	UVLGT	99.19	85.15	91.98	93.22	90.77
MDS-Net + DFE	UVLGT	**99.22**	**85.60**	**92.24**	**93.60**	**90.92**

TABLE VII
ABLATION EXPERIMENTS OF EACH COMPONENT OF THE MDS-NET ON THE LEVIR-CD+ DATASET

Method	Decoder	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
Baseline + Sub	CSWT	98.73	74.43	85.34	86.19	84.31
Baseline + DFE	CSWT	98.75	74.58	85.44	86.9	84.03
MDS-Net + Sub	UVLGT	98.75	74.77	85.56	86.84	84.33
MDS-Net + DFE	UVLGT	**98.78**	**75.25**	**85.88**	**86.90**	**84.88**

TABLE VIII
ABLATION EXPERIMENTS OF EACH COMPONENT OF THE MDS-NET ON THE WHUCD DATASET

Method	Decoder	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
Baseline + Sub	CSWT	99.47	88.08	93.66	95.34	92.04
Baseline + DFE	CSWT	99.51	88.91	94.13	96.36	92.01
MDS-Net + Sub	UVLGT	99.50	88.67	93.99	95.62	92.42
MDS-Net + DFE	UVLGT	**99.54**	**89.55**	**94.49**	**96.38**	**92.66**

TABLE IX
ABLATION EXPERIMENTS OF EACH COMPONENT OF THE MDS-NET ON THE CDD DATASET

Method	Decoder	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
Baseline + Sub	CSWT	98.99	91.83	95.74	95.55	95.93
Baseline + DFE	CSWT	99.13	92.90	96.32	96.29	96.34
MDS-Net + Sub	UVLGT	99.21	93.75	96.68	96.42	96.94
MDS-Net + DFE	UVLGT	**99.32**	**94.39**	**97.12**	**97.03**	**97.20**

TABLE X
ABLATION EXPERIMENTS OF EACH COMPONENT OF THE MDS-NET ON THE SYSU-CD DATASET

Method	Decoder	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
Baseline + Sub	CSWT	92.27	70.73	82.85	86.89	79.18
Baseline + DFE	CSWT	92.54	71.37	83.30	88.20	78.90
MDS-Net + Sub	UVLGT	92.43	70.91	82.98	88.33	78.25
MDS-Net + DFE	UVLGT	**92.62**	**71.77**	**83.57**	88.02	**79.54**

TABLE XI
FIVE-FOLD CROSS-VALIDATION EXPERIMENTAL RESULTS OF THE SEVEN COMPARISON MODELS ON THE LEVIR-CD DATASET

Model	Backbone	IoU(μ/σ)	F1(μ/σ)	Prec(μ/σ)	Rec(μ/σ)
ChangeFormer[41]	Transformer	82.40/0.066	90.30/0.038	92.09/0.114	88.68/0.105
DMINet[69]	Resnet18	82.58/0.059	90.46/0.035	92.59/0.320	88.43/0.337
GAS-Net[44]	Resnet18	83.60/0.182	91.05/0.584	91.08/0.386	**91.06**/0.106
CF-GCN[8]	Resnet50	83.12/0.165	90.78/0.098	92.52/0.475	89.11/0.299
ChangeCLIP(RN50)[36]	Resnet50	84.87/0.067	91.81/0.038	93.41/0.145	90.27/0.194
MDS-Net(ViT-B/16)	**Swin-T**	**85.54**/0.080	**92.21**/0.049	**93.51**/0.073	90.94/0.123

TABLE XII
COMPARISON OF MDS-NET'S PERFORMANCE ON RSCD DATASETS USING TEXT PROMPTS INFERRED BY THE CLIP MODEL CORRESPONDING TO REMOTE SENSING IMAGES

Dataset	Backbone (Image encoder)	CLIP Initialization (Text encoder)	OA(%)	IoU(%)	F1(%)	Prec(%)	Rec(%)
LEVIR-CD	Swin-T	Resnet50	99.21	85.36	92.10	93.35	90.89
		ViT-B/16	**99.22**	**85.60**	**92.24**	**93.60**	**90.92**
LEVIR-CD+	Swin-T	Resnet50	**98.87**	**76.81**	**86.89**	**88.51**	**85.32**
		ViT-B/16	98.78	75.25	85.88	86.9	84.88
WHUCD	Swin-T	Resnet50	99.51	88.99	94.18	95.99	92.43
		ViT-B/16	**99.54**	**89.55**	**94.49**	**96.38**	**92.66**
CDD	Swin-T	Resnet50	**99.33**	**94.51**	**97.18**	**97.03**	**97.32**
		ViT-B/16	99.32	94.39	97.12	97.03	97.20
SYSU-CD	Swin-T	Resnet50	92.33	70.50	82.70	**88.33**	77.73
		ViT-B/16	**92.62**	**71.77**	**83.57**	88.02	**79.54**

visual representations within the decoder, outperforming CSWT across all metrics in the five datasets. MDS-Net demonstrates superior performance compared to the single-modal model pretrained on ImageNet through language-guided fine-tuning with context-aware prompting. These promising results indicate that our model can significantly enhance the performance of vision models in downstream RSCD tasks.

In summary, our comprehensive ablation analysis quantitatively demonstrates the unique contributions of the proposed multimodal encoder, DFE module, and UVLGT multimodal decoder in MDS-Net. The experimental results validate the innovations of our approach.

G. Randomized Performance Evaluation

For the system-level comparison, we present the mean (μ) and standard deviation (σ) from a five-fold cross-validation experiment to reduce the impact of randomness. After the original LEVIR-CD dataset was scrambled, the numbers of slices in the training, validation, and test sets are also 7120, 1024, and 2048, respectively, resulting in a total of five datasets. From Table I–V, we selected several well-performing models and conducted multiple randomized independent experiments on the five datasets. Table XI shows the five-fold cross-validation experimental results on the LEVIR-CD test set. The experimental results further confirm the reliability and stability of the proposed method, with no signs of model overfitting observed.

H. Comparison of MDS-Net Using Different Pretrained CLIP Models

We employ the text prompts inferred by the CLIP's ResNet50 (RN50) and ViT-B/16 (ViT) models to guide the training of MDS-Net. As shown in Table XII, ViT outperforms RN50 in the WHUCD and SYSU-CD datasets, achieves comparable performance to RN50 in the LEVIR-CD dataset, and performs

TABLE XIII
COMPARISON OF THE PERFORMANCE OF DIFFERENT IMAGE MODELS ON THE LEVIR-CD AND WHUCD DATASET

Backbone (Image encoder)	CLIP Initialization (Text encoder)	Differential method	Decoder	LEVIR		WHUCD	
				IoU	F1	IoU	F1
ResNet50	/	Sub	FPN	83.81	91.19	86.55	92.79
ResNet50	/	Sub	CSWT	84.18	91.41	86.86	92.97
Swin-T	/	Sub	CSWT	84.49	91.59	88.08	93.66
ResNet50	Resnet50	DFC	VLDD	84.90	91.83	86.46	92.74
ViT	ViT-B/16	DFC	VLDD	83.41	90.96	86.99	93.04
ResNet50	Resnet50	DFE	UVLGT	85.27	92.05	87.81	93.51
ViT	ViT-B/16	DFE	UVLGT	83.82	91.20	88.70	94.01
Swin-T	Resnet50	DFC	VLDD	85.01	91.90	88.93	94.14
Swin-T	ViT-B/16	DFC	VLDD	85.21	92.02	89.28	94.34
Swin-T	Resnet50	DFE	UVLGT	85.36	92.10	88.99	94.18
Swin-T	ViT-B/16	DFE	UVLGT	**85.60**	**92.24**	**89.55**	**94.49**

worse than RN50 in the LEVIR-CD+ and CDD datasets. While different text encoders will guide the backbone network to exhibit slightly varied performance on different datasets, overall, they all demonstrate high accuracy. These results clearly indicate that the text encoder can successfully guide MDS-Net through language priors, thereby improving performance. The findings presented in this section motivate proposing a solution to extend the applicability of knowledge acquired through large-scale vision language pretraining to a broader spectrum of models. We anticipate that this direction will lead to intriguing advancements in RCSD in the future.

I. Comparison of the Performance of Different Models

CNNs excel in feature extraction and spatial modeling, while Transformers demonstrate superior capabilities in capturing global context and long-range dependencies. As shown in Table XIII, we evaluate the performance of the proposed model by employing ResNet [59], ViT [63], and Swin transformer [11] as backbone networks, along with various differential methods (Sub, DFC [36] and DFE) and decoder structures (FPN [71], CWST [70], VLDD [36], and UVLGT). It is evident that Swin-T outperforms the other backbone networks across both metrics when utilizing Sub/FPN, Sub/CWST, DFC/VLDD, or DFE/UVLGT individually. Furthermore, we observe that with the same backbone network, DFE/UVLGT outperforms Sub/FPN, Sub/CWST, and DFC/VLDD across both metrics, indicating the important role of differential methods and decoders in change detection. These are attributed to the Swin transformer and UVLGT's superior ability to capture global contextual information in remote sensing images.

J. Feature Visualization

To further evaluate MDS-Net's effectiveness in capturing change features for RSCD tasks, we provide heatmap visualizations comparing the baseline model with various MDS-Net components across the LEVIR-CD, LEVIR-CD+, WHUCD, CDD, and SYSU-CD datasets. As shown in Fig. 9, the highlighted areas denote significant correlations between the selected foreground points and pixels of identical semantic categories, where feature attention is concentrated. Compared to the baseline, each component of MDS-Net exhibits a stronger focus on foreground, change, and global information, leading to more accurate target localization and refined boundaries. In addition, we observe that our model more effectively identifies fine-scale and small regions and is capable of detecting unlabeled changes. This highlights the effectiveness of the proposed multimodal vision-language representation learning and differential feature methods in RSCD tasks.

K. Complexity Analysis

Table XIV presents the efficiency comparisons of various models on LEVIR-CD+, WHUCD, CDD, and SYSU-CD. We evaluated the overall computational complexity of the network's encoder and decoder by measuring the number of parameters (Param.) in million (Mb), floating-point operations per second (FLOPs), and frames per second (FPS). MDS-Net's FLOPs are much lower than those of ChangeFormer [41] and GAS-Net [44], its Params is slightly lower than that of ChangeCLIP [36], and its FPS is marginally lower than STADE-CDNet [7]. Nevertheless, MDS-Net achieves the best performance across all five datasets. This can be attributed to the incorporation of advanced multimodal vision-language structures in both the encoder and decoder, which integrate textual semantic information with remote sensing images to enhance the visual model's capability in perceiving changes. Furthermore, the DFE module optimizes the differential feature fusion strategy in change detection, facilitating a more effective learning of differential features.

V. DISCUSSION

A. Adaptability of MDS-Net to Any Visual Backbone

In the experiments, we leveraged the language priors from CLIP pretrained models and the image priors from the Swin transformer pretrained on ImageNet to construct a language-guided multimodal change detection framework. Although the feature maps of the image encoder may not strongly correlate with the text features output by the text encoder, this methodology is based on the assumption that the text encoder can

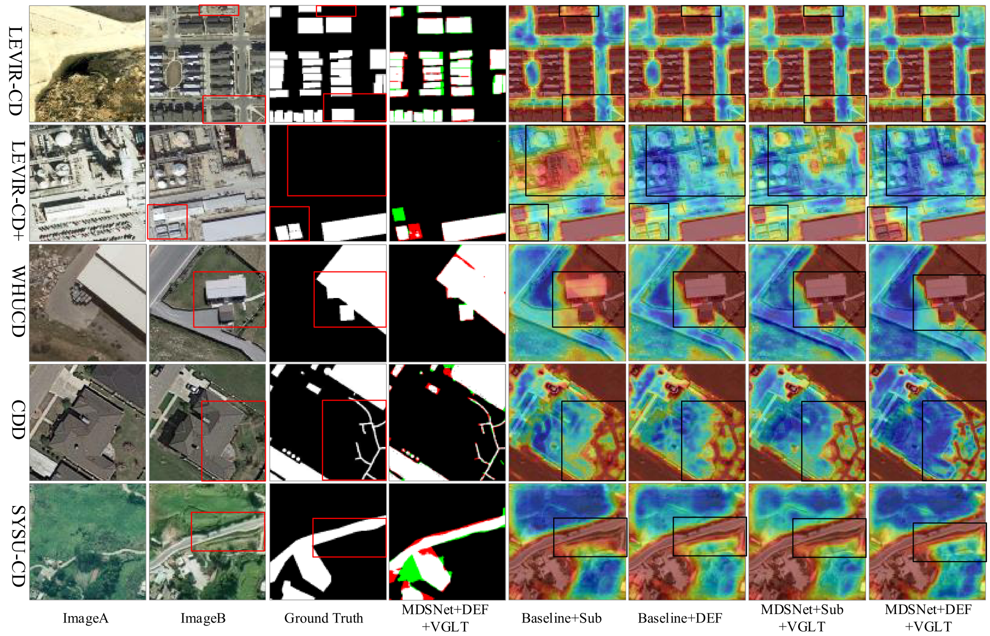

Fig. 9. Comparative visualization of heatmaps between baseline methods and MDS-Net components on the LEVIR-CD, LEVIR-CD+, WHUCD, CDD, and SYSU-CD datasets.

TABLE XIV
EFFICIENCY COMPARISON OF DIFFERENT METHODS IN TERMS OF PARAMETERS, FLOPS, AND FPS

Method	Params.(Mb)	FLOPs(Gbps)	FPS	LEVIR-CD+	WHUCD	CDD	SYSU-CD
				mIoU(%)	mIoU(%)	mIoU(%)	mIoU(%)
BIT[40]	3.5	10.6	67.9	68.86	68.11	76.66	63.09
ChangeFormer[41]	41.0	202.8	31.6	69.62	77.51	89.57	63.40
ICIF-NET[42]	23.8	25.4	22.2	72.37	78.22	91.53	62.27
DMINet[69]	6.3	14.6	110.5	73.44	70.47	93.54	66.21
GAS-Net[44]	23.5	290.0	31.2	65.98	78.37	91.71	62.86
CF-GCN[8]	13.6	44.1	34.6	70.95	80.22	92.49	66.17
STADE-CDNet[7]	3.5	12.0	14.8	69.81	69.53	74.02	57.11
ChangeCLIP[36]	120.8	41.5	22.2	<u>74.81</u>	<u>86.99</u>	<u>93.72</u>	<u>71.10</u>
MDS-Net	117.8	113.7	12.7	**76.81**	**89.55**	**94.51**	**71.77**

effectively guide the image encoder's capabilities in RSCD. The ablation experiments detailed in this paper conclusively demonstrate that integrating visual features with textual features consistently enhances all baseline models across diverse datasets. In addition, we also observe that the ability to capture global contextual information is crucial for improving the accuracy of RSCD. The findings presented in this section provide a solution for generalizing knowledge acquired from large-scale vision language pretraining to broader models. This direction shows promise for bridging vision and language research in the future.

B. Multimodal Vision Language Guided Decoder

In the decoding stage, we introduce a novel mechanism to further improve the change detection performance. This integration enables the model to expand the attention area, effectively capturing complementary information between visual and textual patterns, thus improving feature representation and enhancing RSCD performance. The UVLGT enhances the alignment and interaction between encoded visual-language features and decoded features, enabling deep modeling of long-range dependencies and improving the model's capacity to capture semantic context in remote sensing images. This end-to-end vision language multimodal fusion method enhances its discriminative power in identifying changes in remote sensing images.

C. Effect of Different Feature Extraction Methods

Exploring the correlation between dual-temporal images is crucial for obtaining accurate change maps since temporal information intuitively reflects changes. We construct a novel DFE module that effectively enhances the ability of MDS-Net to learn change features. The experimental results indicate that techniques such as multifeature fusion and differentiation aid networks in capturing change features more effectively. The ablation experiments on the five datasets demonstrate that MDS-Net significantly improves network performance by leveraging the DFE module.

VI. Conclusion

In this article, we introduce MDS-Net, a language-guided multimodal change detection framework that integrates visual contextual information with textual data and utilizes pixel-text score maps to guide the model's learning process. The differential feature enhancement module leverages the complementary information within differential features, enabling the network to adaptively extract and learn change features from target areas. To evaluate the performance of MDS-Net, we conducted a series of experiments on five datasets: LEVIR-CD, LEVIR-CD+, CDD, WHUCD, and SYSU-CD. Our model excels in preserving local detail features, extracting more comprehensive boundary information, and accurately identifying small targets. The experimental results demonstrate that our proposed model achieves SOTA change detection performance compared with other networks. Our research explores how to leverage vision-language multimodal models for change detection, which may provide valuable insights for future studies.

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